Graph Sketching: Polar Curves – Part 2

Questions 1–10

Question 1. Sketch each of the following curves for $0 \leq \theta \leq 2\pi$ which are given in polar form:

a) $r = 2\sec\theta$ where $\theta \notin \{\frac{\pi}{2}, \frac{3\pi}{2}\}$

b) $r = -6\operatorname{cosec}\theta$ where $\theta \notin \{0, \pi, 2\pi\}$

Question 2. Sketch each of the following curves in polar form for $0 \leq \theta \leq 2\pi$. Label the values at which they cross the coordinate axes.

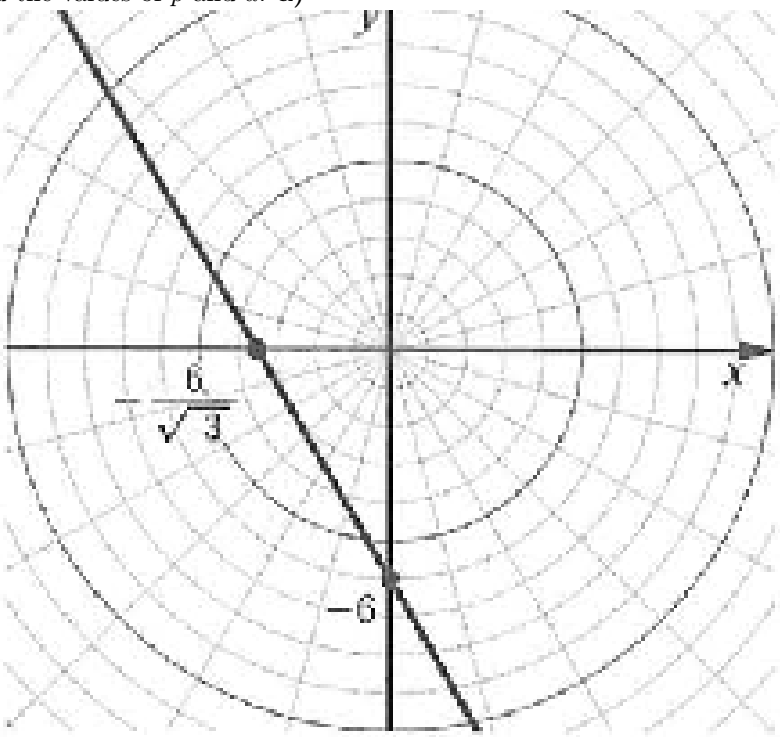
a) $r = 2\sec(\frac{\pi}{4} - \theta)$ where $\theta \notin \{\frac{\pi}{4}, \frac{5\pi}{4}\}$

b) $r = 2\operatorname{cosec}(\frac{\pi}{4} - \theta)$ where $\theta \notin \{\frac{3\pi}{4}, \frac{7\pi}{4}\}$

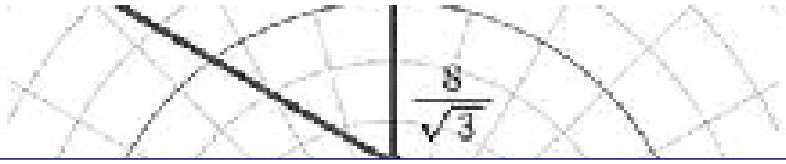
c) $r = -3\operatorname{cosec}(\frac{\pi}{6} - \theta)$ where $\theta \notin \{\frac{\pi}{6}, \frac{7\pi}{6}\}$

d) $r = -4\sec(\frac{5\pi}{6} - \theta)$ where $\theta \notin \{\frac{\pi}{3}, \frac{4\pi}{3}\}$

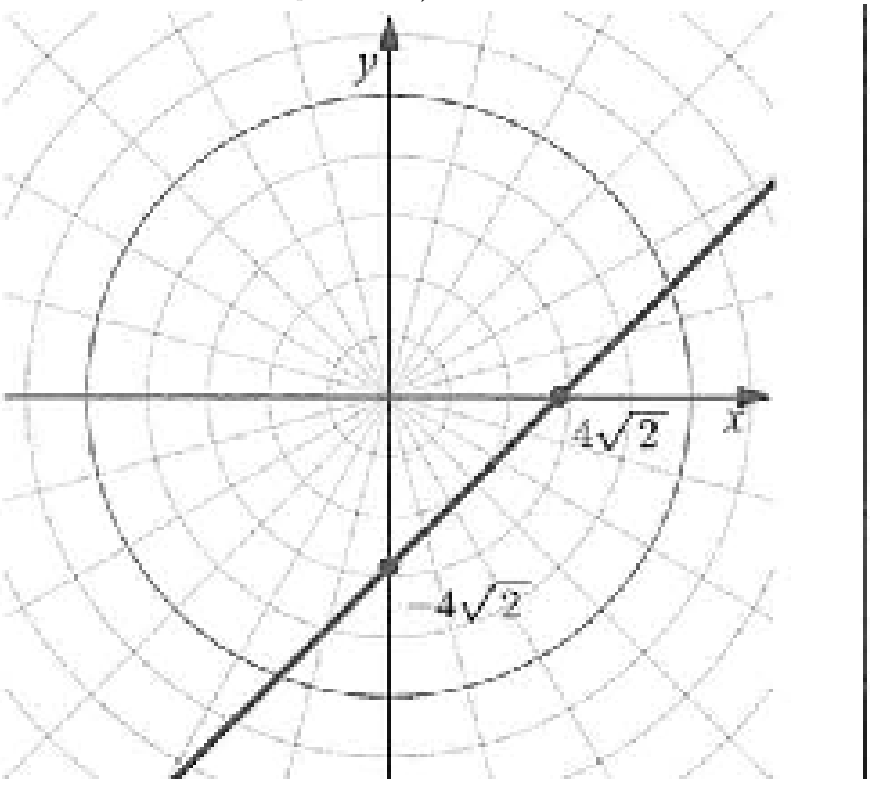
Question 3. The following curves are of the form $r = p\sec(a - \theta)$ for integer p, $0 < a < \pi$ and $0 \leq \theta \leq 2\pi$. In each case find the values of p and a. **a)**



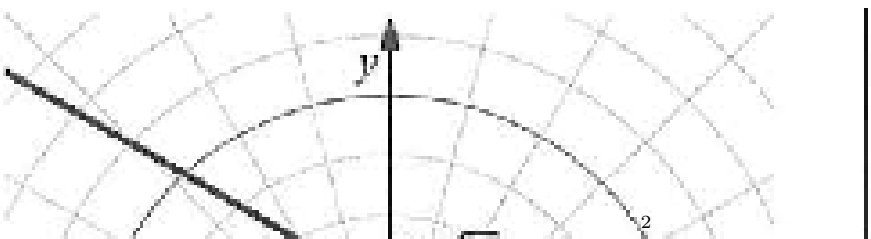
b)



Question 4. The following curves are of the form $r = p\,\mathrm{cosec}(a - \theta)$ for integer p, $0 < a < \pi$ and $0 \leq \theta \leq 2\pi$. In each case find the values of p and a. **a)**



b)



Question 5. Sketch each of the following, given that k is a positive constant, and $0 < \alpha < \frac{\pi}{2}$.

a) $r = k\sec(\alpha - \theta)$

b) $r = k\,\mathrm{cosec}(\alpha - \theta)$

Question 6. Sketch $r^2 = 16\cos 2\theta$ for each of the following restrictions:

a) $0 \leq \theta \leq \frac{\pi}{4}$

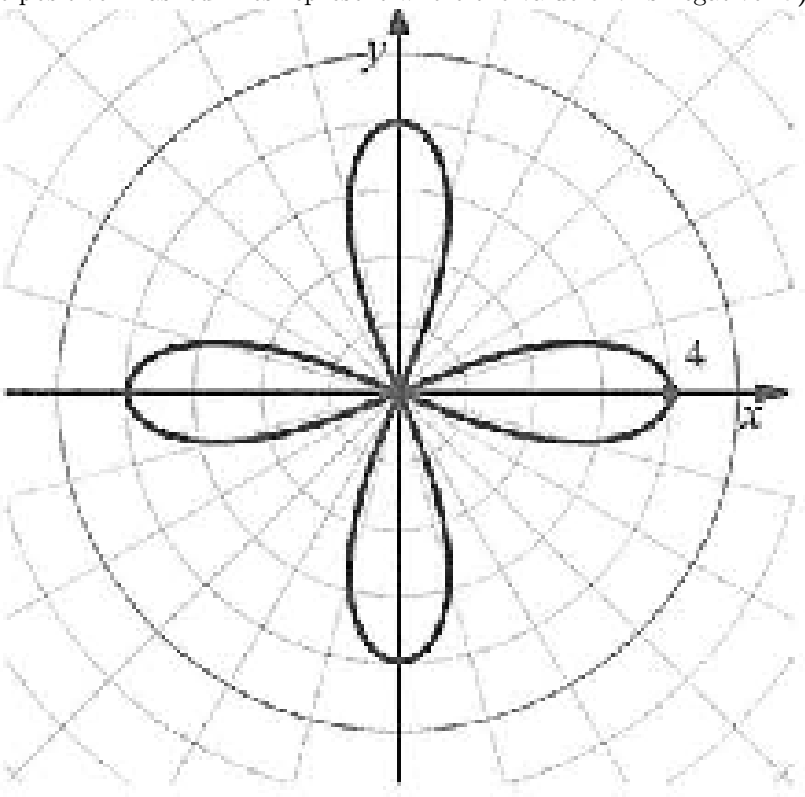
b) $\frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}$

c) $\frac{7\pi}{4} \leq \theta \leq 2\pi$

Question 7. Sketch each of the following curves which are given in polar form. You may assume that θ is restricted to only values for which the curve exists. In each case label a point which lies furthest from the pole with its value of r:

a) $r^2 = 25\sin 2\theta$
b) $r^2 = 10\cos 3\theta$
c) $r^2 = 12\sin 4\theta$
d) $r^2 = 9\sin 6\theta$
e) $r^2 = 14\cos 5\theta$
f) $r^2 = 7\sin 5\theta$

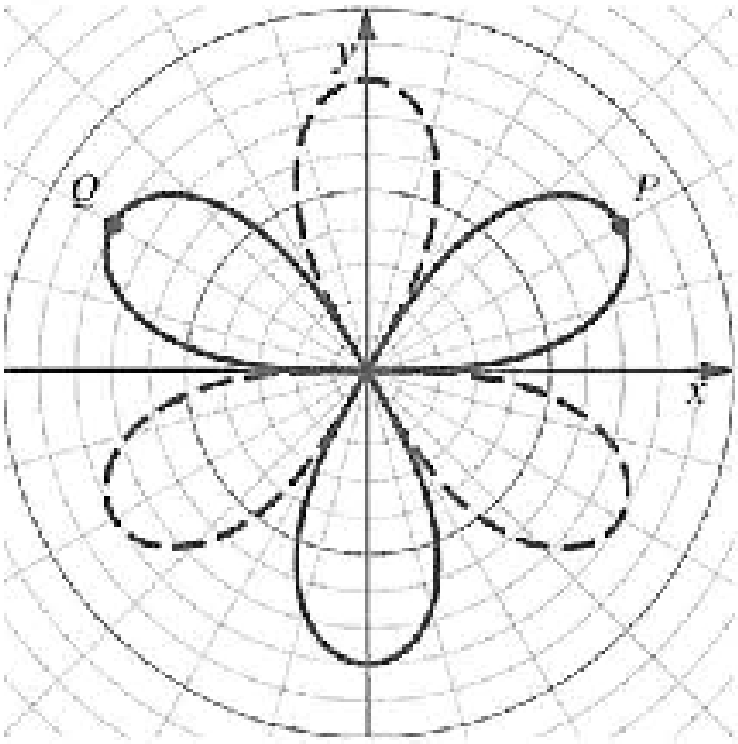
Question 8. Find the polar curve which are equations of the form $r^2 = a\cos p\theta$ or $r^2 = b\sin q\theta$ where a, b, p and q are positive. Dashed lines represent where the value of r is negative. **a)**



b)



Question 9. The curve C has polar equation $r^2 = 64 \sin 3\theta$ and is shown on the graph below.



a) Find the Cartesian coordinates of the point P.
b) Find the Cartesian coordinates of the point Q.
c) Find the points of intersection of the curve C with a second curve given by $r = \sqrt{32}$ for $-\pi < \theta \leq \pi$ giving your

Question 10. The curve C_1 is defined by $r^2 = 16 \cos 4\theta$ for $0 < \theta < \pi$.
a) Determine the values of θ in the interval $0 < \theta < \pi$ for which the curve exists.
A second curve C_2 is given by $r = 4 \cos 4\theta$ for $0 < \theta < 2\pi$.
b) Describe two similarities and two differences between the curves C_1 and C_2.